ASSIGNMENT-1

TOPIC: BASICS OF GEOSPATIAL ANALYSIS

COURSE NAME:

GEOSPATIAL ANALYSIS FOR ENVIRONMENTAL STUDIES

COURSE CODE: ECCSS 4

SUBMITTED TO:

DR. PRAVEEN KUMAR

SUBMITTED BY:

ANSHUMAN SINGH

ROLL NO.: M2024ECCSS003

INDEX

Sno.	Title	Page no.
1.	Data Acquisition and Management	3
2.	Basic Preprocessing	5
3.	NDVI Calculation	7
4.	Supervised Classification	9
5.	Inferences	11

Software used: Arc Map 10.7.1.

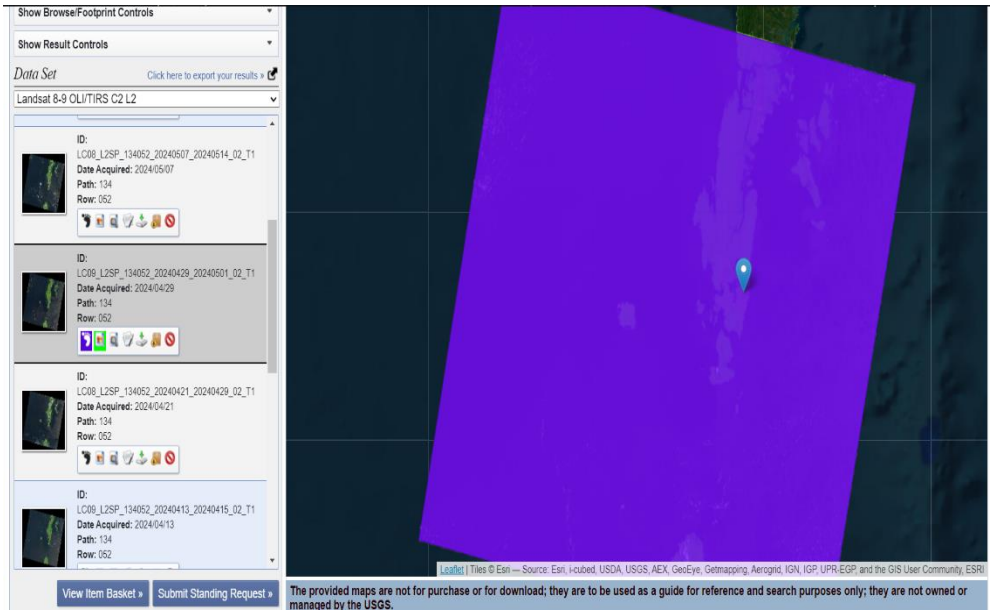
State/Union Territory: Andaman and Nicobar Islands

District/Region selected: South Andaman

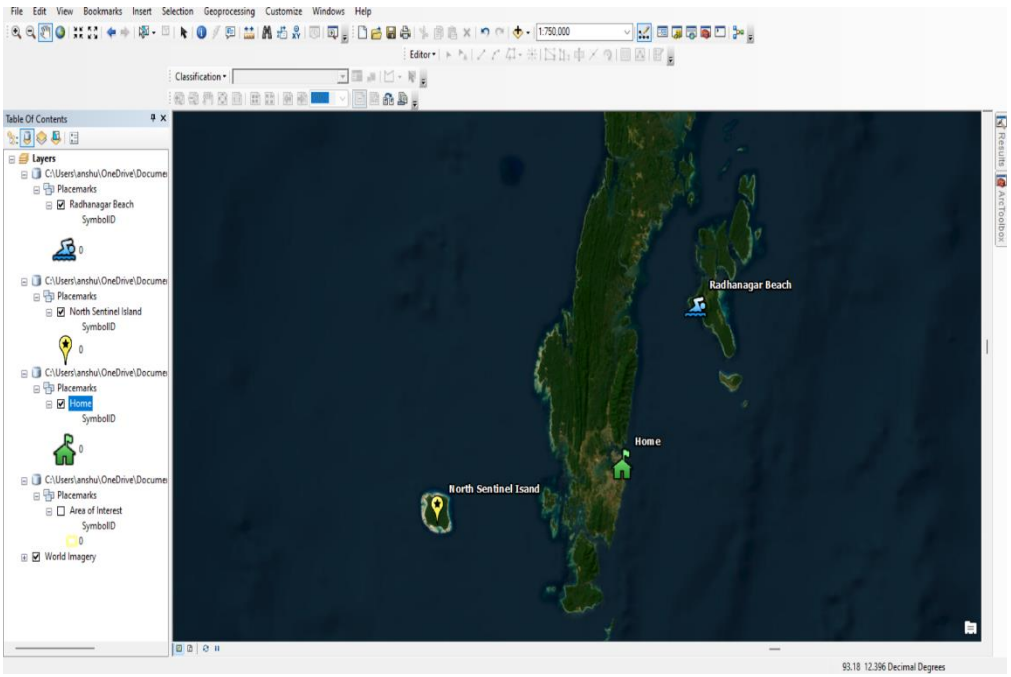
Area of Interest: Sri Vijaya Puram (formerly Port Blair)

1. Data Acquisition and Management

The data was downloaded from ***USGS Earth Explorer***, and the data is from the ***Landsat 8***.

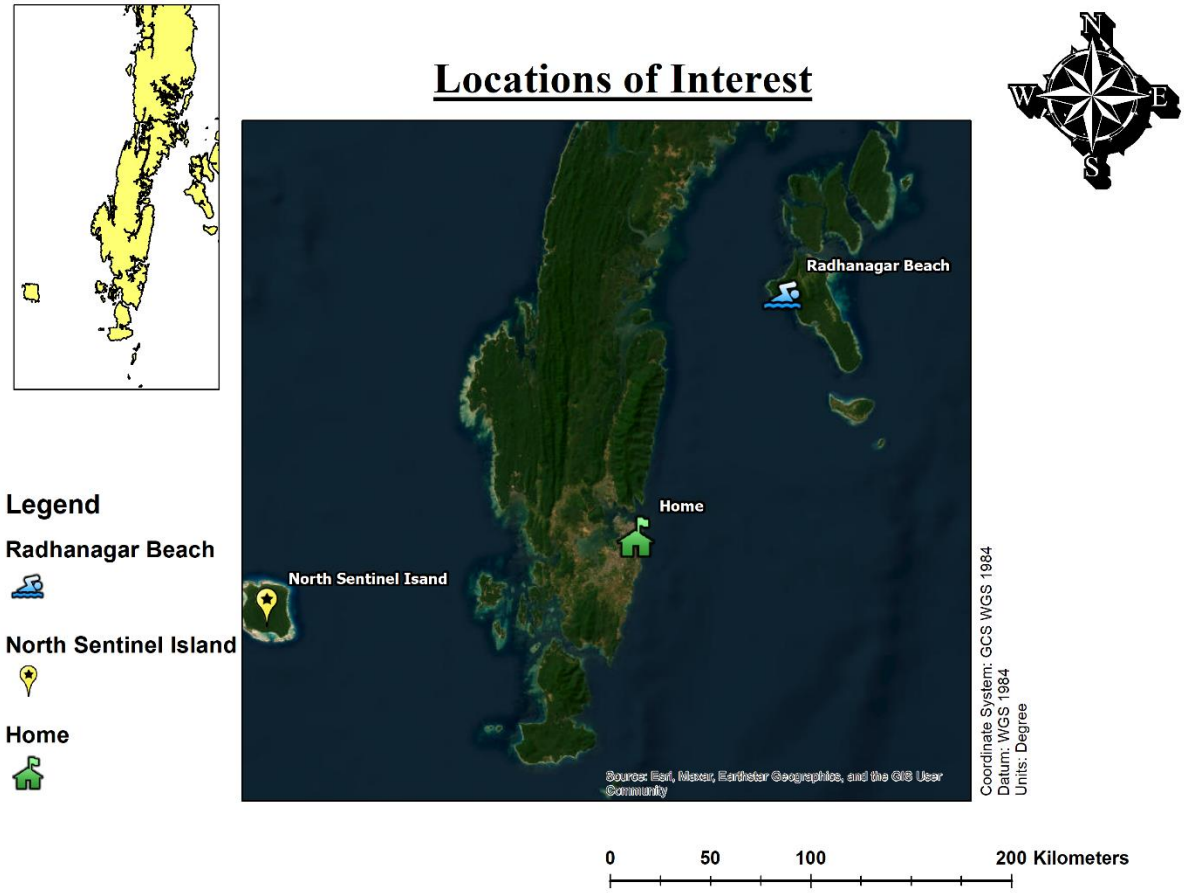


<u>Downloading of Landsat data for South Andaman Islands</u>



Three locations of interest were taken on Google Earth Pro and exported as ".kmz" files. The ".kmz" files were then imported on Arc Map 10.7.1 as layers by the Conversion tool, "from kml", in the Arc toolbox. A basemap was used from the Arc Map's "Add Basemap feature", ***World Imagery***.

- **OUTPUT 1:**



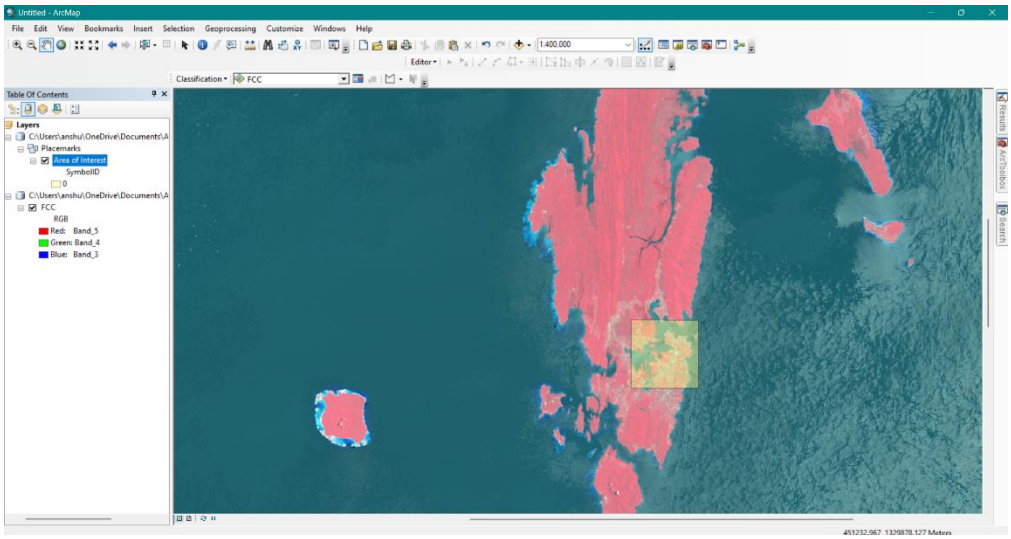
Map 1- Locations of Interest

The locations selected are:

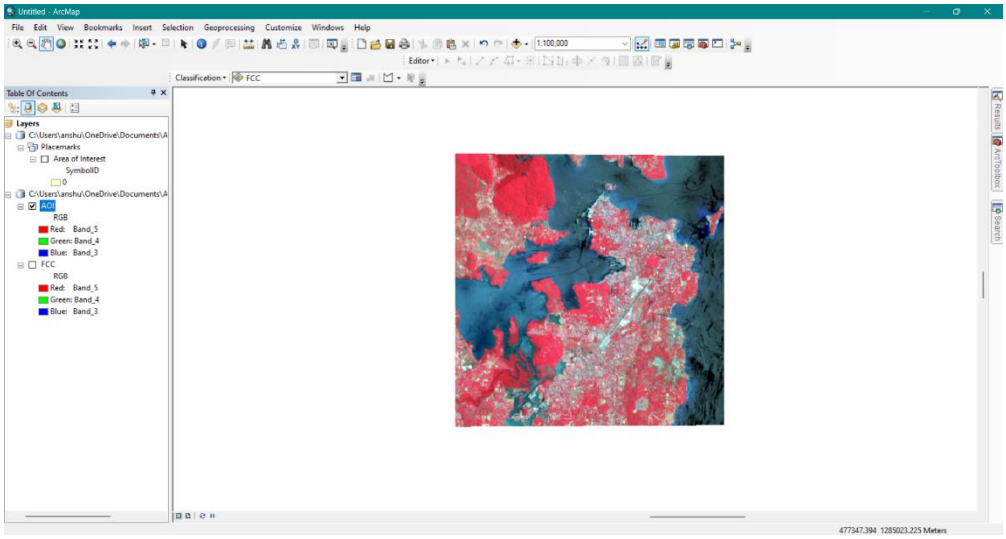
i. Radhanagar Beach
ii. North Sentinel Island
iii. My Home

2. Basic Preprocessing

The Landsat data was required to be clipped to a smaller ***Area of Interest*** for further processing and analysis. The bands (1 to 7) were opened and were made into composite bands from Arc toolbox (*Data management- Raster- Raster Processing- Composite Bands*). The image was converted into ***False Colour Composite*** by changing the band composition to ***5,4,3***.



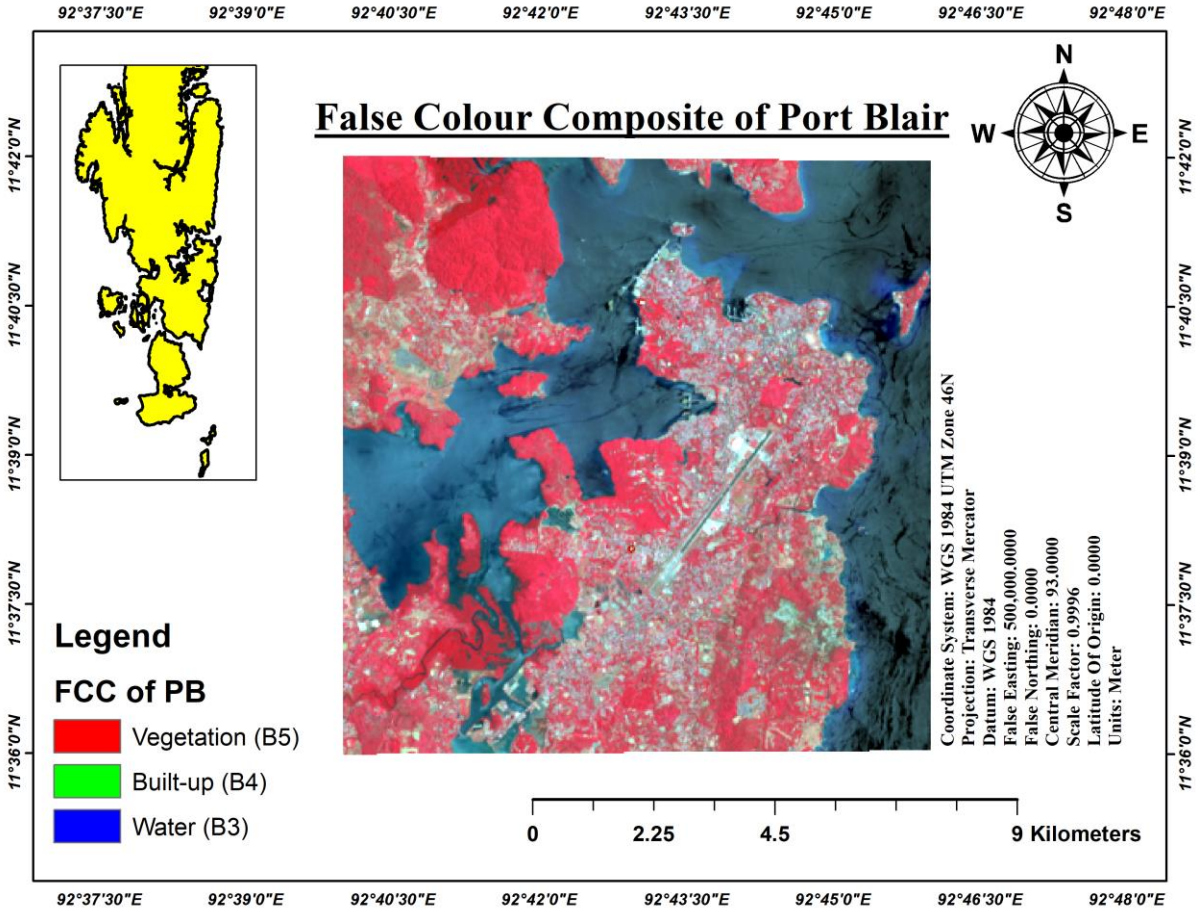
Area to be clipped over Landsat image of South Andaman Islands



Clipped image of Port Blair

Initially the image came out in the True Colour Composite with a band composition of 1,2,3 which was converted into False Colour Composite to have better distinction between the features in the image.

- **OUTPUT 2:**



Map 2-False Colour Composite of Port Blair

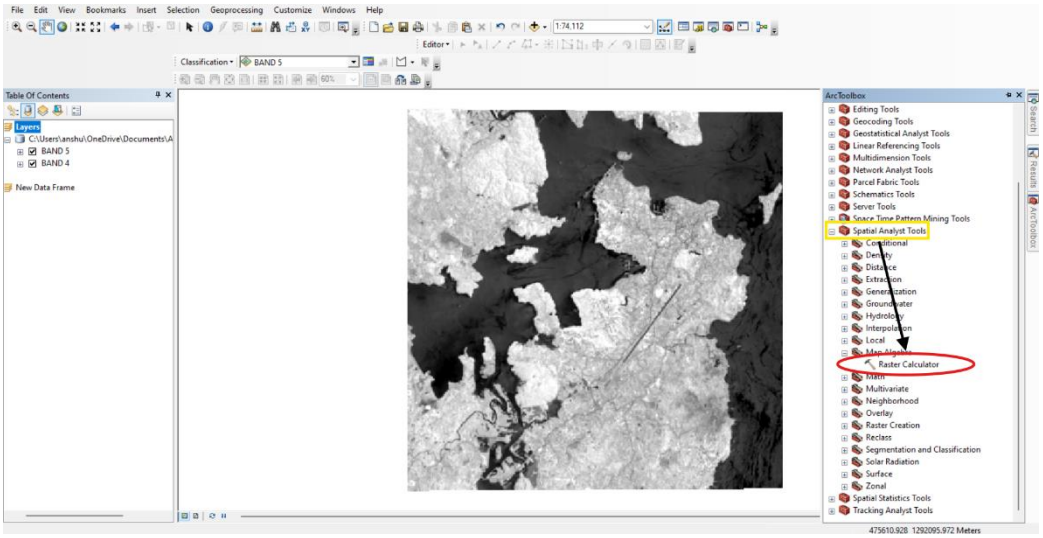
3. NDVI Calculation

For the Normalised Difference Vegetation Index calculation, only the ***Red Band*** (0.64 - 0.67 µm) and ***Near Infrared Band*** (0.85 - 0.88 µm) are required and in terms of Landsat 8 data they are ***Bands 4 & 5***.

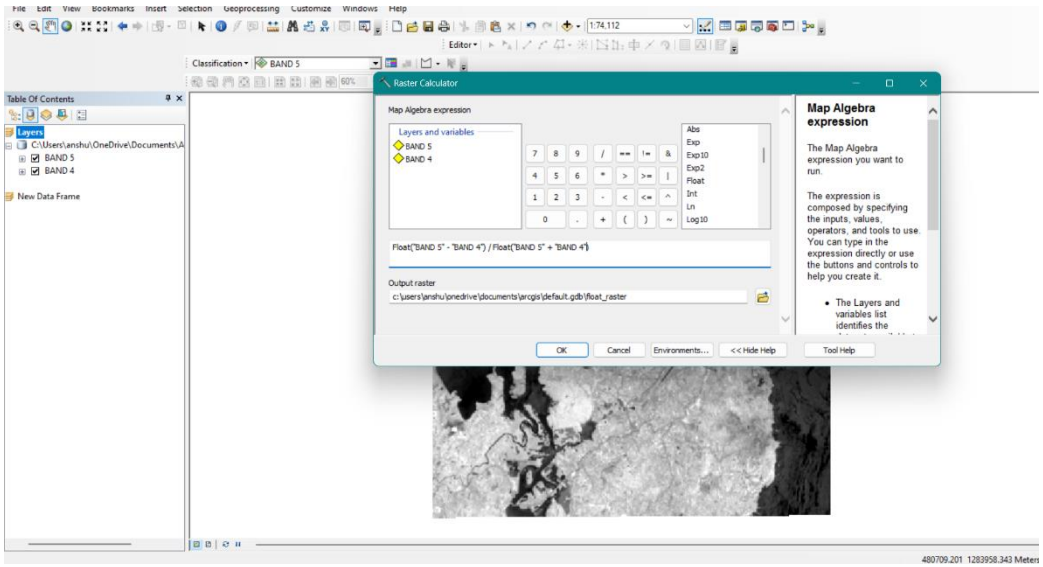
<u>FORMULA</u>:

$$NDVI=(NIR+RED)/(NIR-RED)$$

$$NDVI=[Band\ 5+Band\ 4]/[Band\ 5-Band\ 4]$$

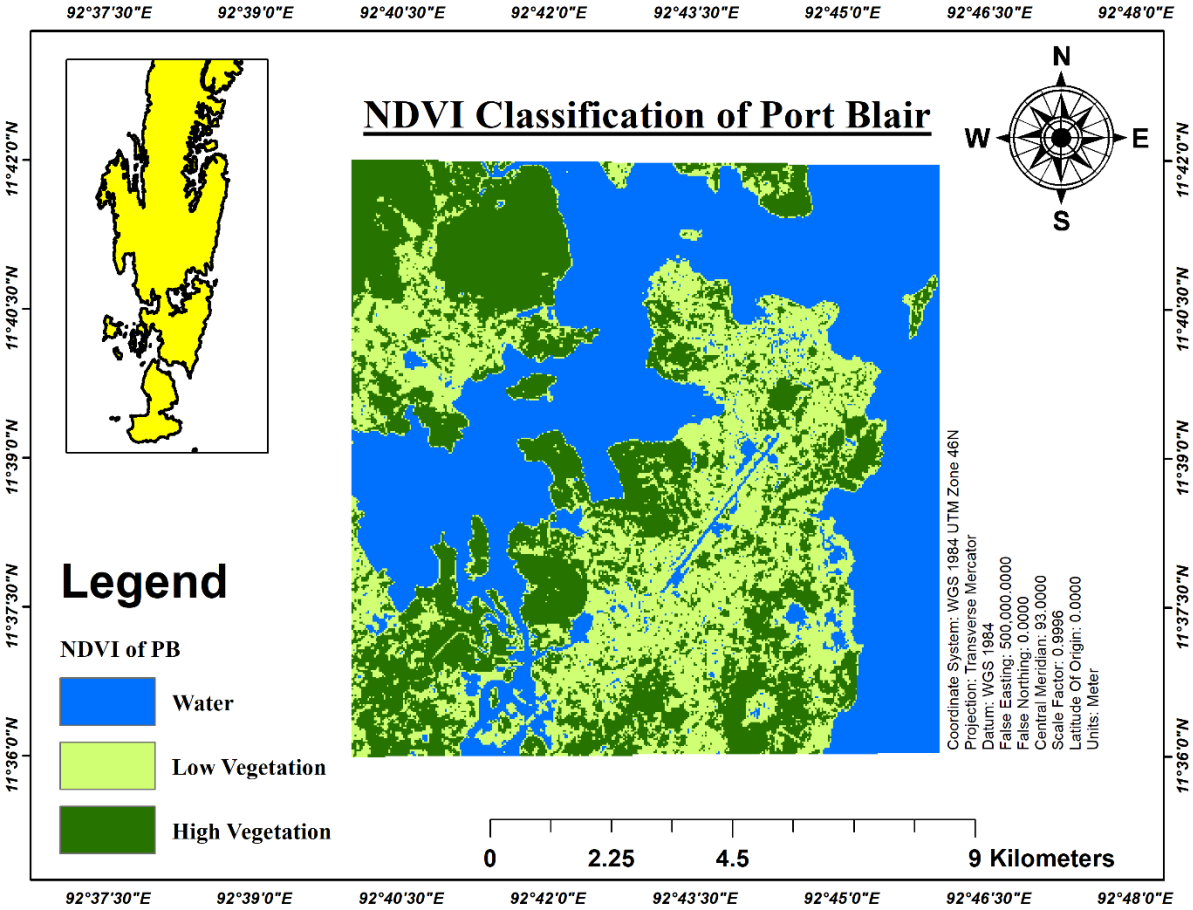


<u>NDVI Calculation procedure</u>



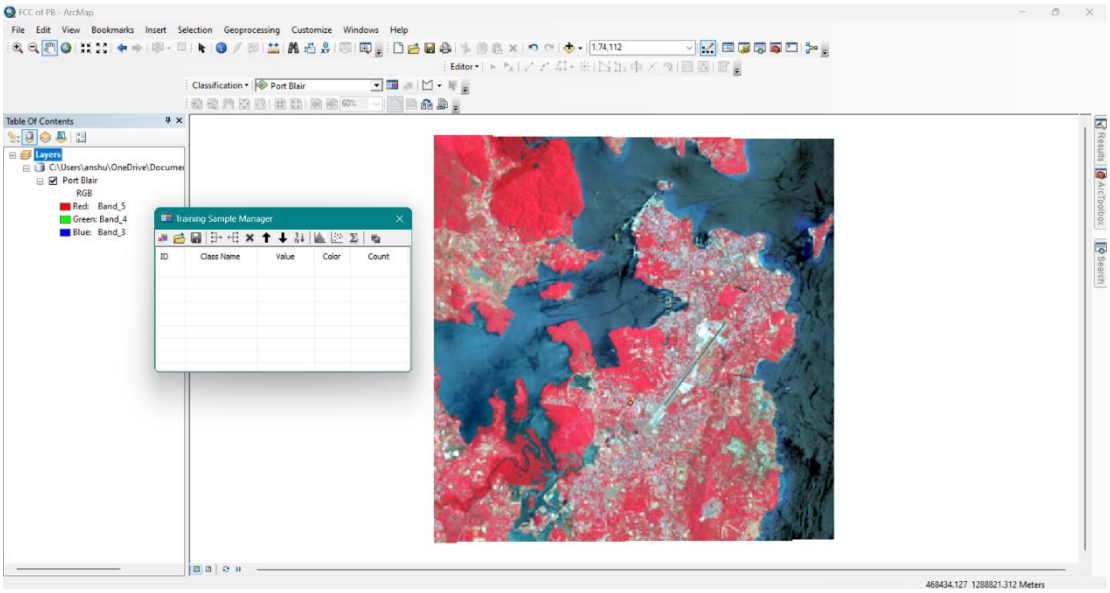
<u>Formula for NDVI calculation in Arc Map</u>

- **OUTPUT 3:**

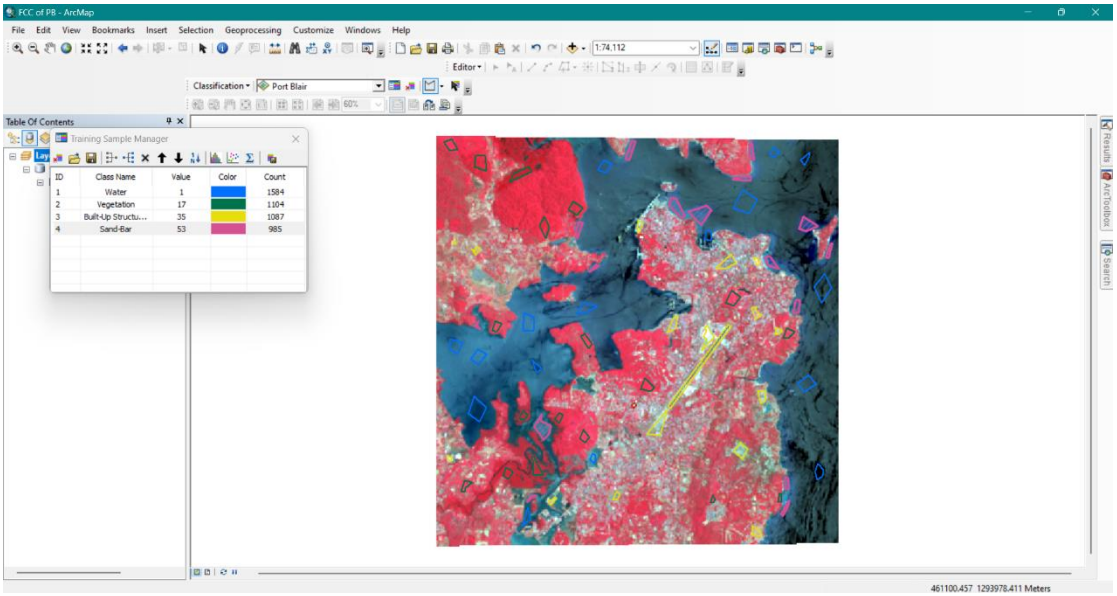


Map 3-NDVI Calculation of Port Blair

4. Supervised Classification

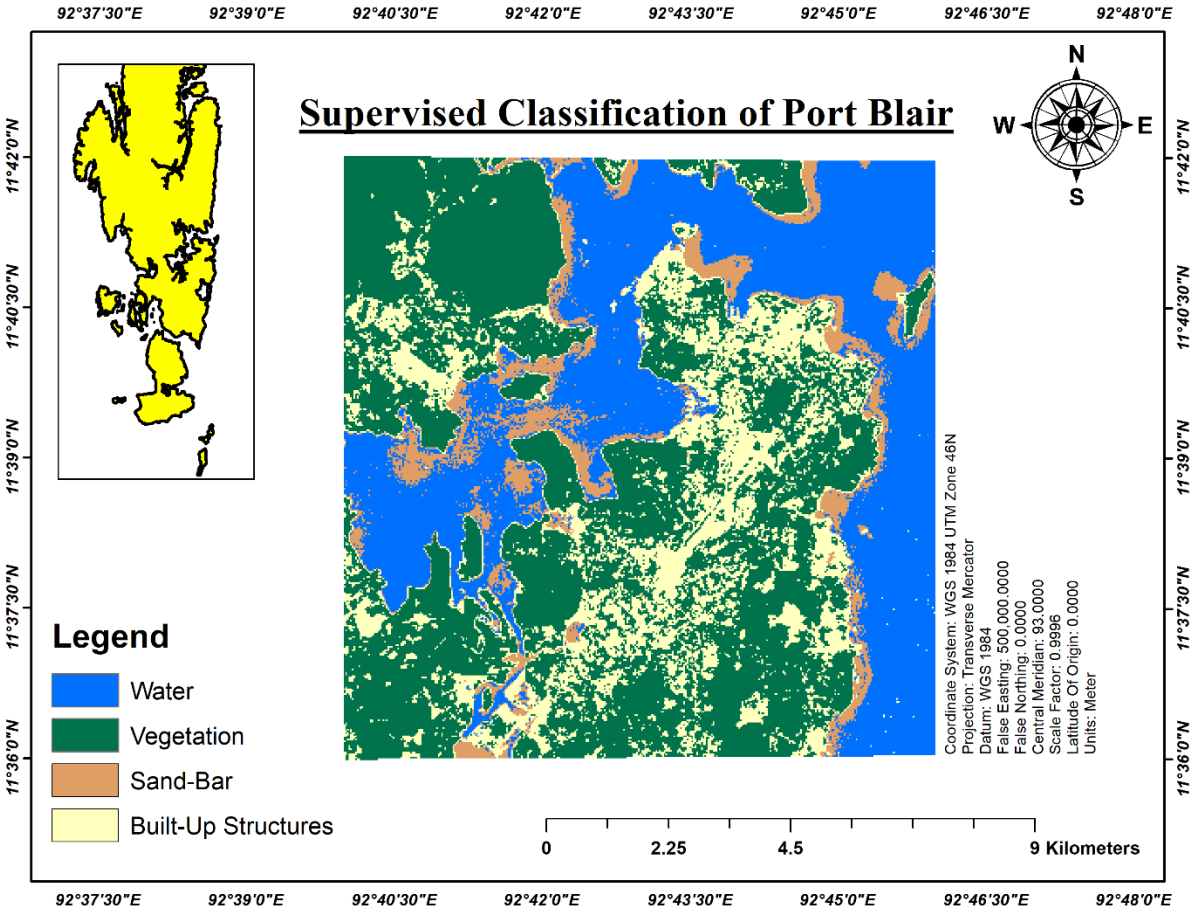


Making of training samples



Merging of training samples for Supervised Classification

- **Output 4:**



Map 4- Supervised Classification of Port Blair

5. Inferences

1. Data Acquisition and Management:

- When going for downloading data from USGS Earth Explorer, knowing about the extent of the cloud cover is very important to get good quality data i.e. less than 5%.
- The use of Arc Map worked good for me but the open-source capabilities of QGIS along with its user-friendly interface makes it a solid option.
- The use of a basemap from Arc Map was a necessity for getting a better output.

2. Basic Preprocessing:

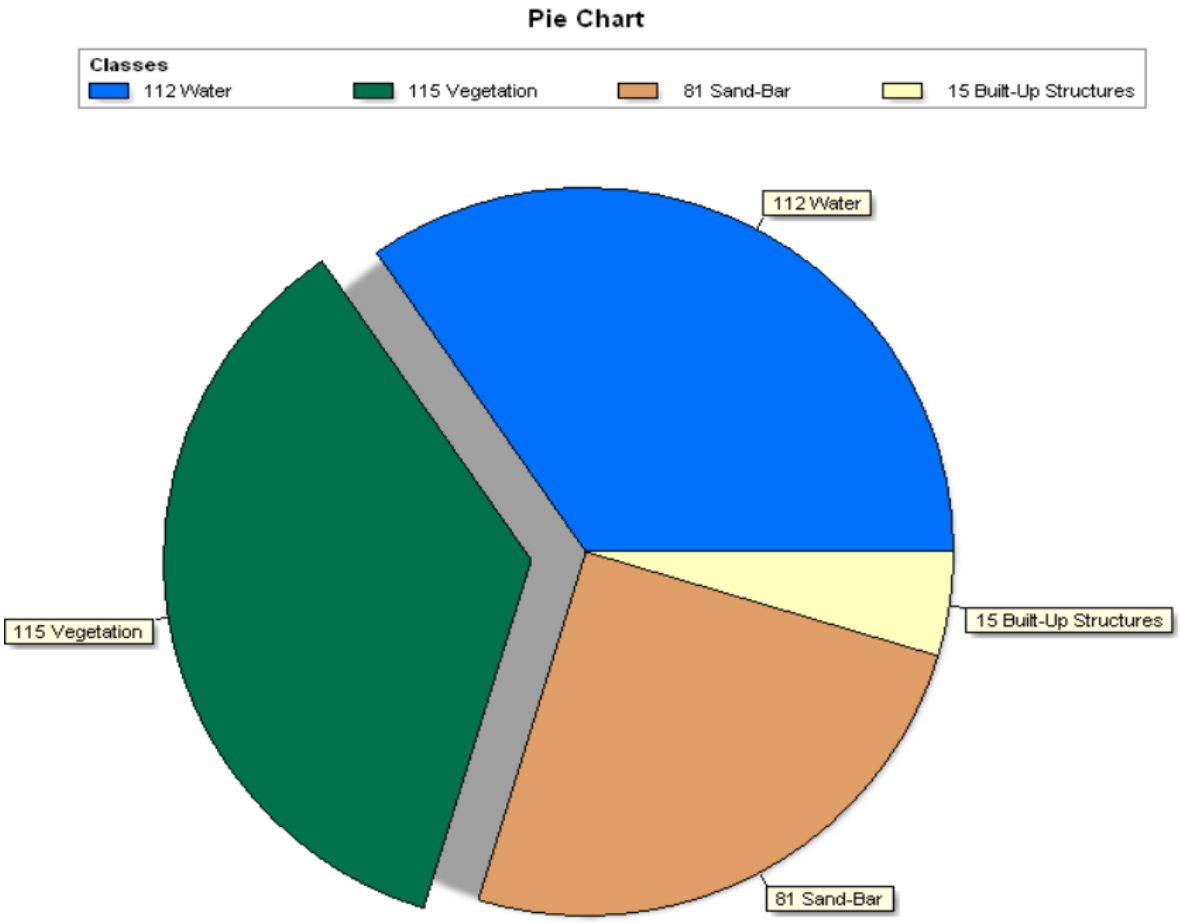
- The area selected by me, Andaman Islands, has varied and unique geographical features. This caused the analysis and processing a bit more complex.
- The clipping of the area is a very good option of identifying the target area that enables a more thorough analysis.
- The process of False Colour Composite conversion makes the distinction of features easier due to the merging of multiple bands and their data.

3. NDVI Calculation:

- Due to the poor quality of the data, the NDVI obtained had some errors.
- Identification of the right bands and the correct formula is key for calculating the NDVI for an area.
- NDVI allows there to be the identification of the vegetation which in the case of my study area is quite significant as it shows the potential to map and track the vegetation cover changes in the islands.

4. Supervised Classification:

- Poor quality led to errors in the classification of the area.
- There was a need of taking a huge number of classes for vegetation and built-up structures to get correct results due to their close proximities.
- The sand-bar class is unique to the islands.



Pie chart representation of Supervised Classification